Radial Poynting flux in capacitors, wires & coax

How circuits truly transport energy through empty space around wires, exact Q29 gap flux proof, and coaxial TEM guides.

Charging capacitor: inward radial energy flux (Q29)

Take circular vacuum plates of radius a, spacing d << a, E = E z_hat and azimuthal B_phi(a) = mu0 Ic / (2 pi a) phi_hat at cylindrical rim r = a:

$$\mathbf{S}(a) = \frac{E\hat{\mathbf{z}} \times B_\phi(a)\hat{\boldsymbol{\phi}}}{\mu_0} = -\frac{EI_c}{2\pi a}\hat{\mathbf{r}}$$

$$\mathcal{P}_{\text{in}} = |\mathbf{S}(a)|\,(2\pi a d) = (Ed)\,I_c = VI_c = \frac{d}{dt}\left(\frac{q^2}{2C}\right)$$

WHY / Notice z_hat x phi_hat = -r_hat! Electrostatic energy stored in the gap enters sideways from the surrounding field across the rim area 2 pi a d.

Cylindrical Ohmic wire: radial inflow feeds Joule heat

A steady DC current I runs along +z through a cylindrical wire of radius a, length l, resistance R = V/I: inside E = (V/l) z_hat and surface B_phi(a) = mu0 I/(2 pi a):

$$\mathbf{S}(a) = -\frac{VI}{2\pi a\ell}\hat{\mathbf{r}} \Rightarrow \mathcal{P}_{\text{in}} = |\mathbf{S}(a)|(2\pi a\ell) = VI = I^2R$$

WHY / Battery fields outside the metal funnel energy radially inward through the wire surface at rate I^2 R, where collisions convert it into lattice heat!

Inside radius r < a, E(r) = V/l is uniform while B_phi(r) ~ r, so inward power across radius r is V I (r^2/a^2) — proportional to enclosed volume!

Coaxial cable: lossless TEM wave power transport

Inner core radius a (+I along +z, potential +V) and outer shield radius b (-I, potential 0) carry radial E_r and azimuthal B_phi in insulator (eps, mu):

$$E_r = \frac{V}{r\ln(b/a)}, \qquad B_\phi = \frac{\mu I}{2\pi r}, \qquad S_z = \frac{VI}{2\pi r^2\ln(b/a)}$$

$$\int_a^b S_z\,(2\pi r\,dr) = VI, \qquad Z_{\text{coax}} = \frac{V}{I} = \frac{1}{2\pi}\sqrt{\frac{\mu}{\varepsilon}}\ln\left(\frac{b}{a}\right)$$

WHY / Integrating S_z across the annular dielectric gives exact circuit power V I! For vacuum coax, Z_coax = 60 ln(b/a) Ohm (50 Ohm standard at b/a = 2.30).

Hidden magnetic-energy term & validity fence

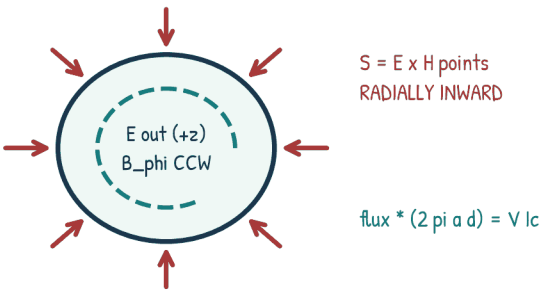
In Q29 with non-steady current q(t) = alpha t^2 (Ic = 2 alpha t), changing B_phi inside the gap induces a small second-order radial correction to E(r):

$$U_B = \int_0^a \frac{B_\phi^2}{2\mu_0}\,2\pi r d\,dr = \frac{\mu_0 I_c^2 d}{16\pi}, \qquad \frac{U_B}{U_E} \sim \left(\frac{\omega a}{c}\right)^2$$

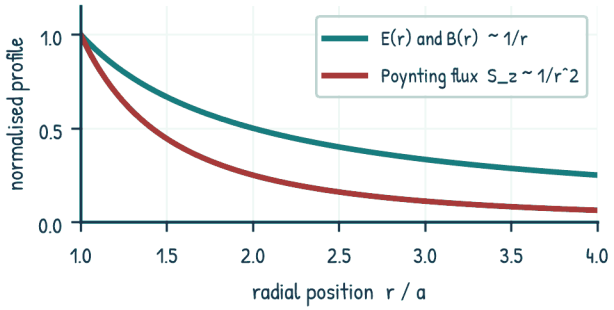
WHY / When variation time tau >> a/c (w a << c), U_B << U_E justifies Model Q. Retention of dU_B/dt gives the next term in the cylindrical cavity expansion!

WATCH / Never extrapolate quasistatic gap fields to t -> 0 shock steps where dI/dt is impulsive: sharp wavefronts travel radially inward at speed c!

Charging capacitor top view: field energy enters radially through the cylindrical rim



Coaxial TEM line: fields fall as 1/r between conductors a and b, carrying total V I



REMEMBER / Wires guide fields, while energy travels in the surrounding space via S = E x H: inward into capacitors/resistors and axially along coax.

Curved-surface forces, cones & stellar beta blow-out

Exact integrals proving F_black = F_specular = I pi a^2/c on a sphere, cone/cylinder drag factors, and astrophysical radiation blow-out.

Why black & specular spheres feel identical axial force

Divide the sunlit hemisphere of radius a >> lambda into ring zones dA = 2 pi a^2 sin(theta) d(theta) at normal angle theta in [0, pi/2] (sec. 2.11, Q14):

$$F_{\text{abs}} = \frac{I}{c}\int_0^{\pi/2} \cos\theta\,(2\pi a^2 \sin\theta\,d\theta) = \frac{I\pi a^2}{c}$$

$$F_{\text{spec}} = \int_0^{\pi/2} \left(\frac{2I}{c}\cos^2\theta\right)\cos\theta\,(2\pi a^2 \sin\theta\,d\theta) = \frac{4\pi a^2 I}{c}\left[\frac{1}{4}\right] = \frac{I\pi a^2}{c}$$

WHY / Physical miracle: a ray hitting at theta scatters at angle pi - 2 theta from forward, carrying away axial momentum -(dU/c)cos(2 theta) whose hemisphere average vanishes!

Cones, cylinders & retroreflectors compared

Apply the exact same local vector projection dF_z = dF_n cos(theta) + dF_t sin(theta) across other symmetric bodies of projected cross-section A_proj:

$$F_{z,\,\text{cone}} = \frac{I\pi a^2}{c}(1 + R\cos 2\alpha_0) = \frac{I\pi a^2}{c}(1 - R + 2R\sin^2\alpha_0)$$

Here alpha_0 is the cone semi-vertical angle (normal makes theta = pi/2 - alpha_0 with axis). Check limits: alpha_0 = pi/2 (flat disc) -> (1 + R) I pi a^2 / c; sharp needle alpha_0 -> 0 with R = 1 -> 0!

WHY / Specular circular cylinder of radius a, length L perpendicular to beam: int_{-pi/2}^{pi/2} 2 cos^3(theta) d(theta) = 8/3 -> F = (4/3)(2 I a L / c) vs black 2 I a L / c!

Stellar radiation blow-out ratio beta (Q34)

Around a star of luminosity L_* and mass M_*, compare outward radiation force I pi a^2/c on a spherical dust grain (density rho_m, radius a) to inward gravity G M_* m / r^2:

$$\beta \equiv \frac{F_{\text{rad}}}{F_{\text{grav}}} = \frac{3L_\star}{16\pi c G M_\star \rho_m a}, \qquad a_{\text{crit}}(\beta = 1) = \frac{3L_\star}{16\pi c G M_\star \rho_m}$$

WHY / Both I = L_*/(4 pi r^2) and gravity fall strictly as 1/r^2, so orbital distance r CANCELS COMPLETELY! For Sun (L_sun = 3.83e26 W, M_sun = 1.99e30 kg) & silicates, a_crit ~ 0.2 um.

WATCH / Do not let beta -> infinity as a -> 0: once grain radius a ~ lambda (~0.5 um), geometric optics breaks down and Mie/Rayleigh scattering cross-section drops as a^6!

Lambertian diffuse reflector: third classic case

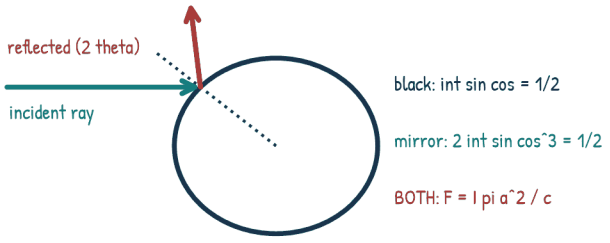
If an opaque flat disc (projected normal incidence) re-scatters all incident energy diffusely with Lambert cosine radiance (~ cos psi into hemisphere), outgoing normal momentum adds 2/3:

$$P_{\text{Lambert, normal}} = \left(1 + \tfrac{2}{3}R_{\text{diff}}\right)\frac{I}{c}$$

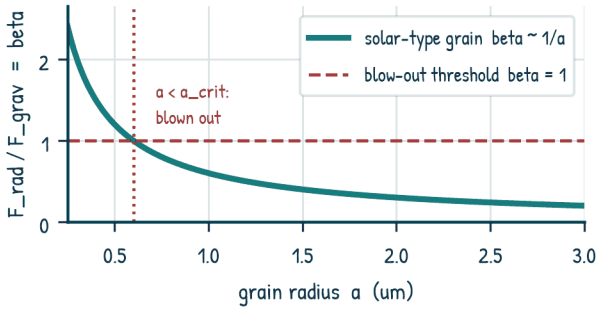
WHY / Diffuse bounce always sits strictly between total absorption (1 * I/c) and normal specular mirror (2 * I/c) because oblique scattered rays waste sideways momentum.

Exemplar check (E8): a = 1.00 mm grain, rho_m = 3000 kg/m^3 at I = 1200 W/m^2 gives F = 12.6 pN and a = 3 I/(4 rho_m a c) = 1.00e-6 m/s^2.

Illuminated sphere zone: local specular scatter equals full black-sphere drag



Radiation-to-gravity ratio beta cancels distance r^2 and scales inversely with radius a



REMEMBER / A specular sphere scatters light into all directions with <cos(2 theta)> = 0, feeling the EXACT same force I pi a^2 / c as a black sphere!

Wave impedance, Fresnel matching & quarter-wave coats

Boundary loops for tangential E and H, why t_E > 1 never violates energy conservation, and impedance-matching layers.

Tangential boundary equations & reflected H sign

Shrink rectangular Faraday and Ampere loops across a charge/current-free planar interface z = 0 between lossless media Z1 (z < 0) and Z2 (z > 0):

$$E_i + E_r = E_t, \qquad \frac{E_i - E_r}{Z_1} = \frac{E_t}{Z_2}$$

WHY / Keeping +x_hat for E_i, E_r, E_t means forward wave has H_i = +E_i/Z1 y_hat, while reflected wave moving along -z_hat has H_r = -E_r/Z1 y_hat!

$$r_E = \frac{Z_2 - Z_1}{Z_2 + Z_1} = \frac{n_1 - n_2}{n_1 + n_2} \ (\mu_1 = \mu_2), \qquad t_E = \frac{2Z_2}{Z_1 + Z_2}$$

Energy conservation & the "t_E > 1" paradox (Q16, Q36)

Intensity is I = E0^2 / (2 Z), so transmitted intensity carries factor 1/Z2 while incident carries 1/Z1. Dividing I_r and I_t by I_i proves exact flux balance:

$$R = r_E^2 = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1}\right)^2, \qquad T = \frac{Z_1}{Z_2} t_E^2 = \frac{4Z_1 Z_2}{(Z_1 + Z_2)^2}, \qquad R + T = 1$$

Paper Q36: for Z2 = 2 Z1, r_E = 1/3 and t_E = 4/3 > 1! Yet R = 1/9 and T = (1/2)(16/9) = 8/9 -> R + T = 1! Larger E amplitude inside larger Z carries LESS current H!

WATCH / Writing T = t_E^2 instead of (Z1/Z2) t_E^2 gives T = 16/9 > 1 — an instant violation of energy conservation whenever Z1 != Z2!

Equal speed != zero reflection in magnetic matter (Q36c)

Can a boundary reflect waves even when BOTH media have the exact same wave speed v1 = v2 (same refractive index n1 = n2)? YES, if permeabilities differ!

$$\mu_2 = 2\mu_1, \ \varepsilon_2 = \varepsilon_1/2 \Rightarrow v_2 = v_1, \quad Z_2 = 2Z_1 \Rightarrow R = 1/9$$

WHY / Conversely, choosing mu2/mu1 = eps2/eps1 makes Z2 = Z1 so R = 0 (invisible Stealth/impedance-matched boundary) EVEN THOUGH wave speed drops across the boundary!

Nonmagnetic glass n = 1.5 in air (Z2 = Z0/1.5): r_E = -0.20 (pi phase flip on E!), R = 0.04 (4% reflection per face), T = 0.96.

Quarter-wave anti-reflection transformer

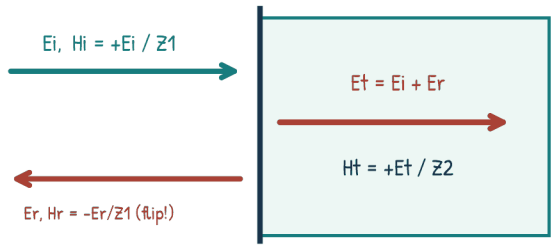
To eliminate reflection completely between media Z1 and Z3 at target vacuum wavelength lambda_0, insert a lossless layer (index n_c, impedance Z_c) of thickness d:

$$d = \frac{\lambda_{coat}}{4} = \frac{\lambda_0}{4n_c}, \qquad Z_c = \sqrt{Z_1 Z_3} \quad (n_c = \sqrt{n_1 n_3} \ \text{ if } \mu_r = 1)$$

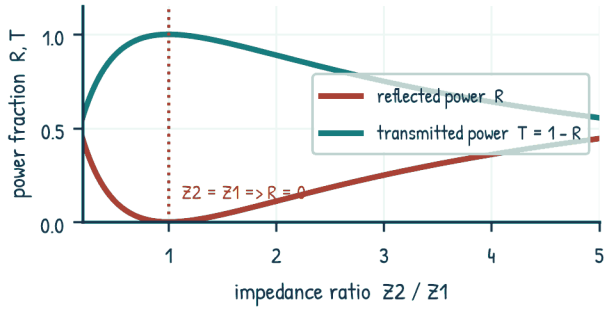
WHY / Equal reflection amplitudes at front and back faces suffer round-trip phase shift 2 k_c d = pi, cancelling destructively in reflection so 100% power transmits!

WATCH / If n1 < n_c < n3, BOTH front and back reflections carry the SAME pi phase flip, so destructive interference still requires 2 n_c d = lambda_0 / 2 (NOT lambda_0)!

Normal-incidence matching: tangential E and H stay continuous; reflected H flips sign



Power reflection R and transmission T vs impedance ratio Z2/Z1 (always R + T = 1)



REMEMBER / Reflection is governed by wave impedance Z = sqrt(mu/eps), NOT refractive index n alone; always compute power transmission as T = (Z1/Z2) t_E^2.

Standing EM waves, mirror stress & cavity pressure u/3

Constructing exact node structure in front of a conductor, magnetic Maxwell stress at a mirror, and 3D thermal photon gas.

Standing wave in front of a perfect mirror (z <= 0)

Superpose E_i = E0 cos(kz - wt) x_hat (+z) and reflected E_r = -E0 cos(kz + wt) x_hat (-z) so tangential E vanishes on mirror z = 0 (Q31):

$$\mathbf{E} = 2E_0 \sin(kz)\sin(\omega t)\,\hat{\mathbf{x}}, \qquad \mathbf{B} = \tfrac{2E_0}{c}\cos(kz)\cos(\omega t)\,\hat{\mathbf{y}}$$

WHY / For E_r moving along -z with -E0 x_hat, (-z_hat) x (-x_hat) = +y_hat: reflected B ADDS at z = 0 instead of cancelling!

Node map: E nodes at z = -m lambda/2 (including mirror z = 0); B nodes at z = -(2m + 1) lambda/4 (quarter-wave shifted!).

Zero mean Poynting vector vs nonzero mirror pressure

Multiply E and B: S_z = (2 E0^2 / Z0) sin(2kz) sin(2wt) sloshes energy back and forth every quarter-cycle with <S_z> = 0, while time-averaged density is spatially uniform:

$$\langle u \rangle = \varepsilon_0 E_0^2 = 2\langle u_{\mathrm{inc}} \rangle, \qquad P_{\mathrm{mirror}} = \left\langle \tfrac{B^2(0,t)}{2\mu_0} \right\rangle = \varepsilon_0 E_0^2 = \tfrac{2I_{\mathrm{inc}}}{c}$$

WHY / At z = 0, E = 0 so surface currents feel Lorentz force (J_s x B_avg_across_sheet) equal to magnetic Maxwell stress B(0,t)^2 / (2 mu0), matching momentum reversal 2 I_inc / c!

WATCH / Never write uE = uB in a standing wave: at wt = 0 ALL energy is magnetic; at wt = pi/2 ALL energy is electric!

Isotropic cavity radiation: why pressure is u / 3

Fill an enclosure uniformly with isotropic radiation of total energy density u: solid angle dOmega carries density du = u dOmega/(4 pi) at angle theta to a specular wall:

$$P_{\mathrm{iso}} = \int_{\theta<\pi/2} \tfrac{2\,du}{1}\cos^2\theta = \tfrac{2u}{4\pi}\int_0^{2\pi} d\phi \int_0^{\pi/2} \cos^2\theta \sin\theta\, d\theta = \tfrac{u}{3}$$

WHY / Only half the rays move toward the wall (factor 1/2), and they strike at all angles with <cos^2(theta)> = 1/3: (1/2) * 2 * (1/3) u = u / 3 (vs u for a collimated beam)!

WATCH / Careful boundary check (sec. 4.3): a cold black wall absorbing an isotropic field of total density u feels u / 6 until thermal equilibrium emission adds the second u / 6!

Adiabatic photon-gas expansion & cavity resonators

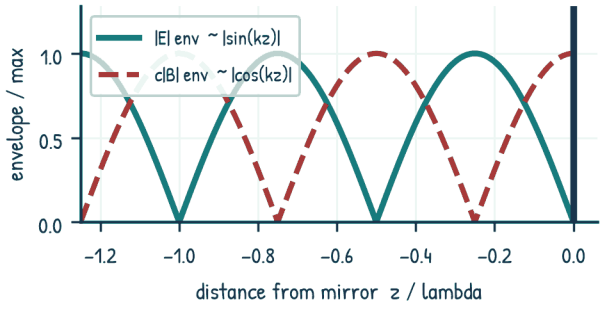
Combining P = u/3 and U = u V inside reversible First Law dU + P dV = T dS = 0 (with u = a T^4) gives the relativistic photon-gas adiabat:

$$PV^{4/3} = \text{const}, \qquad TV^{1/3} = \text{const} \quad (\gamma_{\mathrm{photon}} = \tfrac{4}{3})$$

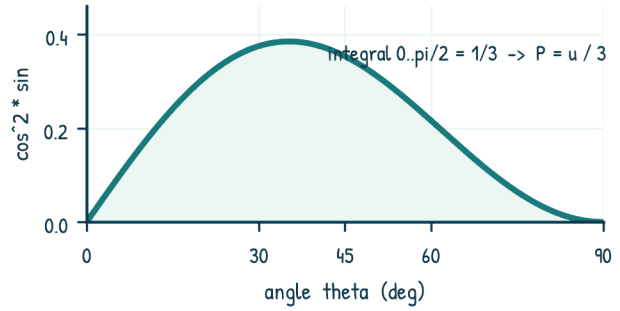
Resonant frequencies of a lossless rectangular metallic box (sides Lx, Ly, Lz):

$$f_{lmn} = \tfrac{c}{2}\sqrt{\left(\tfrac{l}{L_x}\right)^2 + \left(\tfrac{m}{L_y}\right)^2 + \left(\tfrac{n}{L_z}\right)^2}$$

Perfect mirror at z = 0 forces an E node and a B antinode (magnetic stress 2 I / c)



Angular weight cos^2(theta) sin(theta) peaks at 35.3 deg and integrates to 1/3



REMEMBER / In a standing EM wave E and B oscillate 90 deg out of phase in BOTH space and time: <S> = 0, yet mirror magnetic pressure equals 2 I_inc / c.

Conducting media, skin depth & collisionless plasmas

Damped telegrapher wave equation, 45 deg B-to-E phase lag in metals, shielding attenuation, and ionospheric plasma cutoff.

Wave equation in an Ohmic conductor (sec. 4.4)

Insert J_f = sigma E into Maxwell-Ampere inside a homogeneous conductor: curl B = mu sigma E + mu eps dE/dt. Taking curl of Faraday yields:

$$\nabla^2\mathbf{E} = \mu\sigma\frac{\partial\mathbf{E}}{\partial t} + \mu\varepsilon\frac{\partial^2\mathbf{E}}{\partial t^2}, \qquad k^2 = \mu\varepsilon\omega^2 + i\mu\sigma\omega$$

WHY / First time derivative breaks time-reversal symmetry (Ohmic dissipation!), making wavenumber k = k_R + i k_I complex so exp(i k z) = exp(i k_R z) exp(-k_I z) attenuates!

Poor-conductor limit (sigma << w eps, e.g. dry soil / seawater at high f): k_I ~ (sigma / 2) sqrt(mu/eps) = sigma Z / 2 is independent of frequency!

Good-conductor skin depth delta & 45 deg magnetic lag

When conduction current dominates displacement current (sigma >> w eps), k^2 ~ i mu sigma w. Since sqrt(i) = (1 + i)/sqrt(2), k_R = k_I = 1/delta:

$$\delta = \sqrt{\frac{2}{\mu\sigma\omega}}, \qquad Z_{\text{metal}} = \frac{\omega\mu}{k} = \sqrt{\frac{\omega\mu}{2\sigma}}(1-i) = \frac{1-i}{\sigma\delta}$$

WHY / Three huge Olympiad consequences: (1) E decays by 1/e every delta and intensity I decays as exp(-2 z/delta); (2) B lags E by pi/4 (45 deg!); (3) magnetic energy uB >> uE inside metal!

Scale check in copper (sigma ~ 5.8e7 S/m): delta ~ 8.5 mm at 60 Hz, ~ 66 um at 1 MHz, and ~ 2.1 um at 1 GHz!

Collisionless cold plasma dispersion & total reflection

In a neutralised gas of free electrons (number density n_e, mass m_e), Newton law m_e dv/dt = -e E gives J = -n_e e v = (i n_e e^2 / m_e w) E (purely inductive, ZERO heating!):

$$\varepsilon_{\text{eff}}(\omega) = \varepsilon_0\left(1 - \frac{\omega_p^2}{\omega^2}\right), \qquad \omega_p = \sqrt{\frac{n_e e^2}{m_e\varepsilon_0}}, \qquad \omega^2 = \omega_p^2 + c^2k^2$$

WHY / For w < w_p, eps_eff < 0 so k is purely imaginary: incident radio waves suffer 100% lossless reflection off the ionosphere or metals below UV plasma edge!

Phase speed > c vs group speed < c check

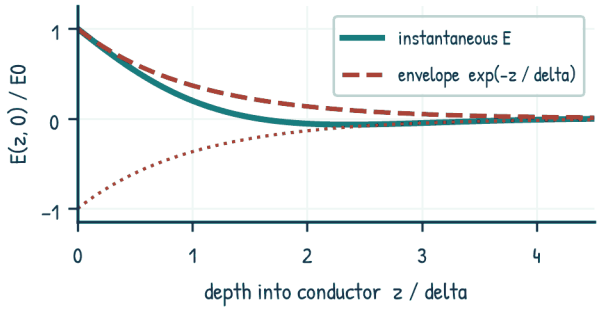
Above plasma frequency (w > w_p), differentiate w^2 = w_p^2 + c^2 k^2 with respect to k to compare crest velocity v_p = w/k with packet/energy velocity v_g = dw/dk:

$$v_p = \frac{c}{\sqrt{1-\omega_p^2/\omega^2}} > c, \qquad v_g = c\sqrt{1-\frac{\omega_p^2}{\omega^2}} < c, \qquad v_p v_g = c^2$$

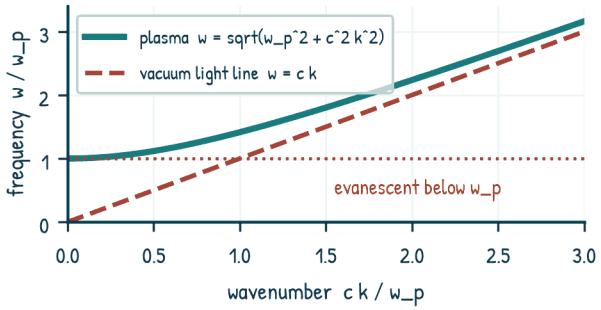
WHY / Relativity forbids signals/energy travelling faster than c — and group/signal velocity v_g stays strictly < c everywhere while crests slide faster along the envelope!

WATCH / In dispersion problems, never plug phase index n = c/v_p < 1 into packet travel delay: radio pulsar dispersion delay uses t_arrival = L / v_g!

Good conductor (k = (1+i)/delta): wave decays by 1/e in one skin depth delta



Collisionless plasma dispersion: v_p > c above cutoff w_p while group speed v_g = c^2/v_p < c



REMEMBER / Good conductors (sigma >> w eps) damp fields exponentially in skin depth delta = sqrt(2/(mu sigma w)); collisionless plasmas reflect below w_p with v_p v_g = c^2 above.

Relativistic Doppler, moving mirrors & waveguides

Exact Lorentz light shift, double boost on moving mirrors, Poynting-Robertson orbital spiral, and parallel-plate TE modes.

Relativistic optical Doppler & transverse time dilation

Because light requires no material medium, source-moving and observer-moving shifts are IDENTICAL and include Lorentz gamma = 1/sqrt(1 - beta^2), beta = u/c:

$$f_{\text{approach}} = f_0\sqrt{\frac{1+\beta}{1-\beta}}, \qquad f_{\text{recede}} = f_0\sqrt{\frac{1-\beta}{1+\beta}}, \qquad f_\perp = \frac{f_0}{\gamma}$$

WHY / Even at closest approach (90 deg in observer frame), clock time dilation slows source oscillations by 1/gamma = sqrt(1 - beta^2) — a pure second-order redshift!

Low-speed check (beta << 1): df / f0 = +- beta + O(beta^2); astronomical redshift parameter 1 + z = lambda_obs / lambda_emit = sqrt((1+beta)/(1-beta)).

Reflection & pressure on a moving flat mirror

A flat perfect mirror moves at speed u = beta c TOWARD an incoming normal vacuum beam of frequency f and intensity I (or minus sign if receding):

$$f' = f\frac{1+\beta}{1-\beta}, \qquad P_{\text{moving mirror}} = \frac{2I}{c}\frac{1+\beta}{1-\beta}$$

WHY / Two clean steps: mirror intercepts crests at relative speed c + u -> rate factor (1 + beta); reflected photon energy also jumps by (1 + beta)/(1 - beta), doing mechanical work on the beam!

Check beta -> -1 (mirror fleeing at light speed): f_ref -> 0 and P -> 0 because photons can never catch the mirror!

Poynting-Robertson drag: why dust spirals INTO stars

A black dust grain of radius a in circular orbit at speed v around a star absorbs radial photons in the star frame, but re-emits thermally isotropically in its OWN rest frame:

$$\mathbf{F}_{\text{PR}} \approx -\frac{I\pi a^2}{c}\frac{\mathbf{v}}{c} = -\frac{L_\star a^2}{4c^2 r^2}\mathbf{v}$$

WHY / In the star frame, forward-beamed thermal emission carries tangential momentum m_eff v = (P_abs/c^2) v away! Or equivalently in grain frame: stellar aberration tilts sunlight slightly head-on by angle v/c!

Result: angular momentum bleeds steadily so surviving micron grains (beta < 1/2) spiral inward over cosmic timescales.

Parallel-plate metallic waveguide (TE_m modes)

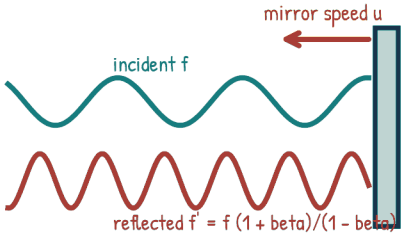
Confine waves between two perfectly conducting plates at y = 0 and y = b: vanishing tangential E_z/E_x on both walls quantises k_y = m pi / b (integer m >= 1 for TE):

$$k_z = \sqrt{\frac{\omega^2}{c^2} - \left(\frac{m\pi}{b}\right)^2}, \qquad \omega_{c,m} = \frac{m\pi c}{b}, \qquad \lambda_{c,m} = \frac{2b}{m}$$

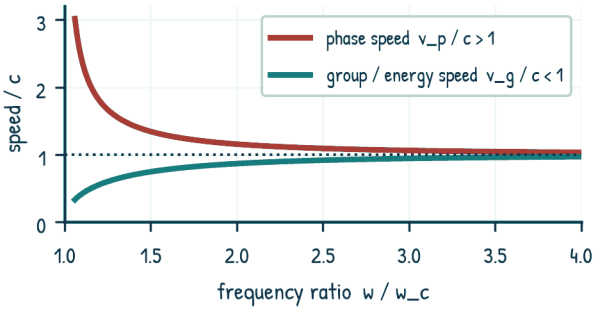
WHY / Picture a plane wave zig-zagging between walls at angle sin(theta_m) = w_c / w: crests slide faster along z (v_p = c / cos theta_m > c) while energy advances at v_g = c cos theta_m < c!

WATCH / A hollow single-conductor pipe has NO TEM mode (m = 0): if lambda_0 > 2 b for parallel plates, ALL TE modes are cutoff and decay exponentially!

Advancing mirror compresses reflected crests twice: double Doppler boost (1+beta)/(1-beta)



Guided TE_1 mode above cutoff w_c = pi c / b: exact product law v_p v_g = c^2



REMEMBER / In vacuum no air frame exists: Doppler depends ONLY on relative beta = u/c, and a moving mirror shifts both frequency and radiation pressure by (1+beta)/(1-beta).